SwinSTFM: Remote Sensing Spatiotemporal Fusion Using Swin Transformer

Guanyu Chen, Peng Jiao, Qing Hu, Linjie Xiao, and Zijian Ye

***Abstract*— Remote sensing images with high temporal and spatial resolutions have broad market demands and various application scenarios. This article aims to generate high-quality remote sensing image time series for feature mining of the growth quality of traditional Chinese medicine. Spatiotemporal fusion is a flexible method that combines two types of satellite images with high temporal resolution or high spatial resolution to generate high-quality remote sensing images. In recent years, many spatiotemporal fusion algorithms have been proposed, and deep learning-based methods show extraordinary talents in this field. However, the current deep learning-based methods have three problems: 1) most algorithms do not support models with large-scale learnable parameters; 2) the model structure based on convolutional neural networks will bring the noise to the image fusion process; and 3) current deep learning-based methods ignore some excellent modules in traditional spatiotemporal fusion algorithms. For the above problems and challenges, this article creatively proposes a new algorithm based on the Swin transformer and the linear spectral mixing theory. The algorithm makes full use of the advantages of the Swin transformer in feature extraction and integrates the unmixing theories into the model based on the self-attention mechanism, which greatly improves the quality of generated images. In the experimental part, the proposed algorithm achieves state-of-the-art results on three well-known public datasets and has been proven effective and reasonable in ablation studies.**

***Index Terms*— Deep learning, remote sensing, spatiotemporal fusion, Swin transformer, unmixing.**

I. Introduction

REMOTE sensing images can be used for crop growth monitoring [1]–[3], land cover changes detection [4], [5], land cover type classification [6]–[8], ecosystem monitoring [9], and other scenes [10], [11], all of which require short revisit periods of satellites and high spatial resolution. The purpose of this article is to enrich the environmental features to better explore the impact of climate and soil conditions on the growth quality of Chinese medicine. This requires the extraction of environmental data, such as humidity based on remote sensing image sequences with high spatiotemporal resolution. Although some recent new satellite systems, such as Sentinel-2, make time series of high-resolution remote sensing images easier to obtain, these images may still have problems such as frequent cloud pollution. At the same time, for some applications involving historical satellite images, it is also necessary to enhance the spatial resolution or temporal resolution of the past remote sensing images. In order to solve these problems, spatiotemporal data fusion algorithms have been proposed in recent years and received extensive attention. This type of algorithm merges satellite images with low spatial resolution but the high frequency (such as MODIS, hereafter referred to as coarse images) and satellite images with high spatial resolution but the low frequency (such as Landsat, hereinafter referred to as fine image) to obtain satellite image time series with both high spatial and temporal resolution.

The current mainstream spatiotemporal data fusion methods can be basically divided into five categories [12]: weight function-based, unmixing-based, Bayesian-based, hybrid-based, and learning-based. Most of these methods require one or more pairs of coarse and fine images and the coarse image at the prediction date as input to output the fine image at the prediction date. Weight function-based methods [13], [14] usually obtain fine pixels by linearly combining the information of the input images. They believe that, when a coarse pixel is "pure" (a coarse pixel only contains one land-cover type), the time difference between coarse images is the same as fine images. Unmixing-based methods [15], [16] consider that a coarse pixel contains multiple land-cover types of fine pixels and uses linear spectral mixing theory to unmix the coarse pixels. In Bayesian-based methods, spatiotemporal fusion is modeled as a maximum posterior probability (MAP) problem. The current Bayesian-based methods [17], [18] mostly complete the prediction of the relationship between the input image and the predicted image by establishing a temporal model and a scale model. And hybrid methods [19], [20] combine the advantages of the above three methods to obtain better performance. Because of its flexibility and great performance, hybrid methods have received extensive attention in the traditional spatiotemporal fusion algorithms.

The abovementioned methods (hereinafter referred to as traditional methods) are basically linear modeling of the remote

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sensing spatiotemporal fusion and have been widely used because of their relatively simple implementation. However, these methods have some shortcomings.

1) In real spatiotemporal fusion scenes, the relationship between images from different sensors and the relationship between images at different dates cannot be simply described by linearity [21]. These linear models may lose some important information about the image and cannot adapt well to images with land-cover changes.
2) In traditional methods, many empirical weight functions are used. Although these functions improve the performance of algorithms, they also bring a large number of manual parameter settings, which limits the generalization of algorithms.

Learning-based methods, especially deep learning-based methods, can well solve the problems of linear models. Thanks to its powerful performance, convolutional neural networks (CNNs) have shined in many fields of computer vision [22]–[24]. At present, more and more spatiotemporal fusion methods use CNN to learn the nonlinear mapping of remote sensing images to improve the performance of algorithms. Due to the similarities between image super-resolution [25]–[27] in computer vision and remote sensing spatiotemporal fusion, some methods [28], [29] have built super-resolution networks to directly increase the spatial resolution of coarse images to the same level as fine images. This kind of method, which directly improves image resolution, could miss a lot of image details because the resolution difference of the remote sensing spatiotemporal fusion scenes (taking Landsat and MODIS as an example, 16 times) is much higher than the image super-resolution (generally two to four times, no more than eight times). Therefore, these algorithms generally add high-pass modulation or linear fusion to correct the generated fine images. Other deep learning-based methods add coarse and fine images at other dates to the networks and complete end-to-end training by fusing more information. They either concatenate the images together to allow the network to complete automatic fine image learning [30]–[33] or learn the nonlinear mapping of input images and then complete the image fusion processes based on the weight functions [34]–[36]. In addition, inspired by image super-resolution, there are also some methods [30], [31], [36], [37] that introduce generative adversarial network (GAN) into the training of spatiotemporal fusion models to enhance the reality of the generated fine images.

Although there have been many deep learning-based spatiotemporal fusion methods in recent years, which have greatly improved the fusion performance, there are still some problems and challenges. First, most of the current end-to-end deep learning-based methods cannot build models with large-scale learnable parameters. Taking GANSTFM [31] as an example, the input images maintain the original resolution during the forward propagation of the network. This can certainly avoid most information loss, but it will also lead to a sharp increase in memory occupation, thus limiting the scale of network parameters. In another type of spatiotemporal fusion method based on image super-resolution, since generating fine images from the coarse image is a process of increasing resolution, the channel dimension can be increased at the lower spatial resolution, thereby increasing the number of network parameters and improving network capabilities. However, due to the lack of spatial details in these algorithms, some postprocessing methods in traditional methods are often used to enhance the result, thus offsetting the increase in image inference speed brought by deep learning methods. Therefore, it is necessary to propose a model that combines end-to-end learning and large-scale network parameters to ensure image inference speed and network learning capabilities at the same time.

Second, CNN-based feature extraction and image reconstruction will bring redundant information. Most deep learning-based spatiotemporal fusion methods use multiple stacked convolutional layers to complete fine image inference. This weight-sharing convolution filtering method can extract local features and learn the texture information of images. However, CNN is essentially a static calculation, and it always maintains the same weight matrix for different feature maps. For the pixel-level task of spatiotemporal fusion, the relationship between pixels in local areas is quite different, and directly using the same convolution kernel to transform and merge them will bring bad effects. For example, in an area, there are two land-cover types: woodland and farmland. If the pixel features of woodland and farmland are merged into farmland pixels by the same transformation, it will inevitably bring the noise to feature information of farmland. In contrast, transformer [38]–[40] will perform dynamic relationship modeling on the input feature maps to enhance feature learning. MSNet [41] uses the transformer encoder to learn the time difference between coarse images, and it only imitates ViT [39] to model the relationship between image blocks. Although this kind of model structure can learn some information from the global image, it does not learn pixel-level attention and cannot benefit from the fine features of neighboring pixels. Therefore, a model is needed to creatively integrate the pixel-level attention mechanism into the spatiotemporal fusion method.

Third, the current deep learning-based methods basically ignore the influence of mixed pixels. Traditional algorithms generally focus on three aspects: eliminating sensor bias, adapting to land cover changes, and processing mixed pixels. In the past, deep learning-based methods have also paid a lot of effort into the first two problems, but none of them have proposed a solution to mixed pixels. BiaSTF [34] is regarded as the closest deep learning-based method to traditional methods, but it only stays at the level of predicting image time differences and sensor bias. Some other nonend-to-end methods [28], [29] also refer to the weight function-based method in the postprocessing module and do not solve mixed pixels well. In summary, a model that combines nonlinear feature learning in deep learning with spatiotemporal fusion ideas in traditional methods should be proposed to benefit from large-scale learnable parameters and excellent theoretical foundations.

To address the abovementioned problems, an end-to-end deep learning method called Swin SpatioTemporal Fusion Model (SwinSTFM) is proposed in this article. The proposed method first extracts nonlinear texture features from the input images based on the shifted window and self-attention mechanism in the Swin transformer [40] and then performs

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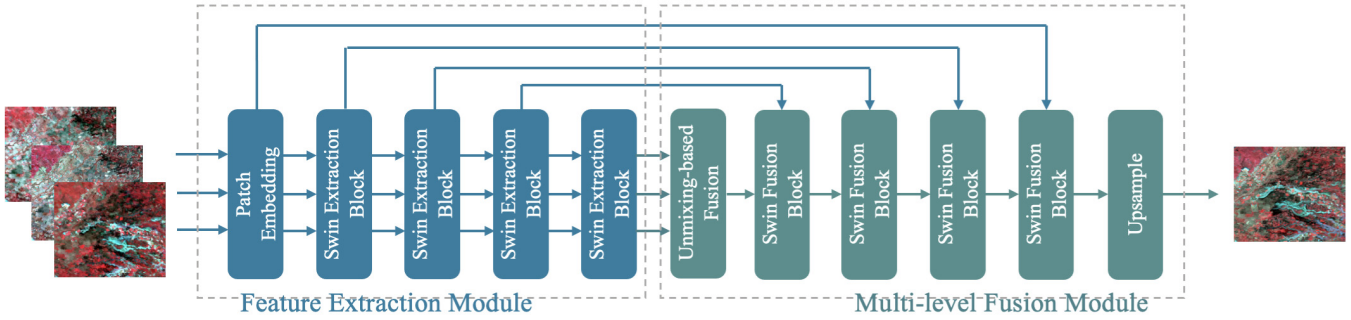


Fig. 1. Architecture of SwinSTFM. Three input images will first go through a patch merging layer for preliminary feature extraction, and then, the image features will be sent to multiple Swin extraction blocks to complete feature optimization. In the MFM, image features first pass through an unmixing-based fusion layer to obtain high-level fine features, and then, the feature fusion of different levels will be completed in four Swin fusion blocks. Finally, the fused features will be sent to the upsample layer to obtain the predicted fine image.

a multilevel fusion of the extracted features based on the attention mechanism and unmixing theories in hybrid-based methods, finally completing the fine image generation at prediction date. SwinSTFM creatively integrates the excellent theories in traditional methods into the deep learning network and has achieved great performance improvements on three public datasets.

In this article, MODIS and Landsat are used as coarse and fine images, respectively. Several excellent algorithms covering traditional methods and deep learning-based methods, including STARFM [13], ESTARFM [14], FSDAF [19], Fit-FC [20], GANSTFM [31], and MSNet [41], are used for comparison. The rest of this article is organized as follows. The theoretical support and network architecture of SwinSTFM are introduced in Section II. The experimental results and comparisons on multiple datasets are presented inSection III. Finally, conclusions are summarized in Section IV.

II. Methodology

SwinSTFM requires one pair of coarse (C_1) and fine (F_1) images at the reference date (t_1) and one coarse image (C_2) at the prediction date (t_2) to predict target fine image (F_2). The overall structure of SwinSTFM is shown in Fig. 1.

SwinSTFM is an end-to-end network composed of a feature extraction module (FEM) and a multilevel fusion module (MFM). It takes three remote sensing images as input and outputs the predicted fine image. The FEM is composed of multiple Swin extraction blocks and extracts features from the input images, respectively. The Swin extraction block is similar to the Swin transformer block. It uses shifted windows and a multihead attention mechanism to achieve feature enhancement and completes image downsampling through multiple blocks. The image features extracted in the FEM will be sent to the MFM to complete feature fusion and fine image prediction. The MFM contains six steps to achieve, of which the first five steps complete the fine feature generation based on the result of the previous block and the image features generated by the Swin extraction block at the same level. Finally, the fine image will be predicted in the sixth step and used to supervise the learning of SwinSTFM with the real image.

A. Feature Extraction Module

As an attention-based architecture, transformer [38] has shown amazing potential in natural language processing tasks, such as sequence modeling and machine translation. The self-attention mechanism in the transformer encoder can model the relationship of the input feature sequence and promote the information fusion of different parts of features. In computer vision, the feature fusion between different regions of the image has been proven to greatly improve the algorithm performance. Therefore, there are also many studies that adapt the transformer encoder structure to multiple tasks in computer vision, and ViT [39] is one of the representative works.

The main structure of ViT is still like the transformer encoder. It first divides the input image into multiple small patches and then uses a linear embedding layer to map each patch into a 1-D vector. At the same time, in order to save the image structure information before mapping, ViT also adds a positional vector for each image patch vector. After that, the network structure is the same as the transformer encoder. These image patches are sent into multiple stacked transformer blocks (as shown in Fig. 2), such as tokens of the input sequence. Then, output refined features based on the multihead self-attention (MSA) module will be used for the following tasks. The calculation of MSA (taking single-head attention as an example) can be given by the following equations:

$$Q = \mathrm{X}\mathrm{W}_Q, K = \mathrm{X}\mathrm{W}_K, V = \mathrm{X}\mathrm{W}_V \quad (1)$$

$$\mathrm{Attention}(Q, K, V) = \mathrm{SoftMax}\left(\frac{QK^T}{\sqrt{d}}\right)V \quad (2)$$

where $X \in \mathbb{R}^{p\times d}$ is the vector matrix of input patches, p is the number of patches, and d is the feature dimension. First, X will be multiplied by three parameter matrices W_Q, W_K, and $W_V \in \mathbb{R}^{d\times d}$ to get query Q, key K, and value V. Then, Q and K will be used to calculate the attention map between patches, and matrix multiplication will be done based on the attention map and V to achieve feature enhancement.

Although ViT has achieved outstanding performance on multiple visual tasks, a very fatal problem of ViT is that the model parameters are too huge, and it needs large amounts of data to pretrain it, which limits the application of ViT on small-scale tasks. Taking remote sensing spatiotemporal fusion as an example, models pretrained on natural image

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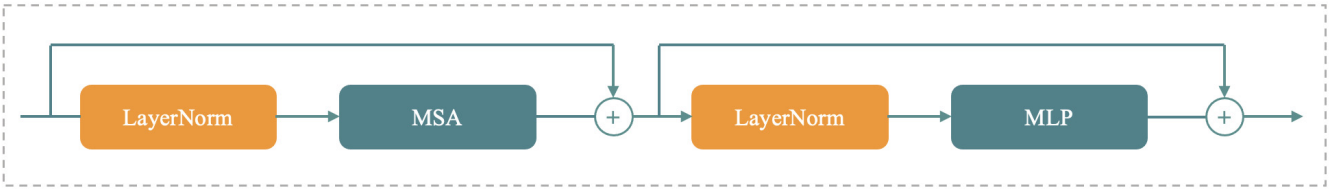


Fig. 2. Architecture of a transformer block.

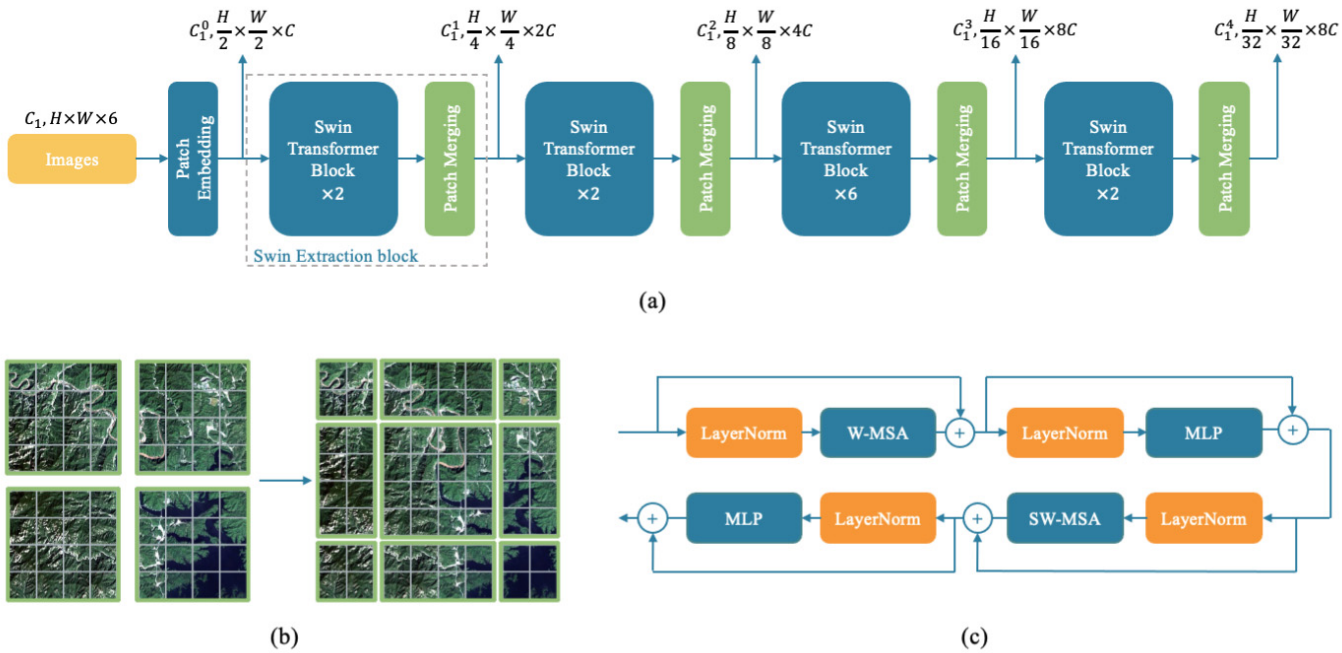


Fig. 3. (a) Architecture of the FEM, taking the coarse image C_1 as an example. C_1^i denotes the coarse feature map generated by the ith block in FEM. (b) Two window partitioning schemes. (Left) Regular window partitioning. (Right) Shifted window partitioning. Images are first divided into several patches (small grids partitioned by gray lines), and then, self-attention is performed within each window (large blocks separated from each other). (c) Two successive Swin transformer blocks. W-MSA and SW-MSA are MSA modules based on two different window partitioning schemes.

datasets, such as ImageNet, are not easy to be transferred to remote sensing image data. At the same time, spatiotemporal fusion datasets are often small-scale because of the difficulty to obtain coarse-fine image pairs, which further limits the application of models with large-scale parameters. In addition, image patches in ViT are relatively large (generally containing hundreds of pixels), which is very unfriendly to pixel-level tasks, such as spatiotemporal fusion, and it may introduce redundant information to features.

In order to use the powerful self-attention module to benefit spatiotemporal fusion, this article uses Swin transformer [40] as the backbone to complete feature extraction. The overall structure of the FEM [as shown in Fig. 3(a)] is basically the same as the Swin transformer with slight differences in some details. First, just like ViT, FEM divides the input remote sensing image into nonoverlapping patches. In order to alleviate the impact of different types of adjacent pixels as much as possible, the size of the image patch in this article is 2×2. Each patch can be regarded as a "token" and transformed into a 1-D vector by a linear mapping layer.

Since traditional spatiotemporal fusion methods need to enumerate all pixels in a local region and complete complex calculations, moving windows are widely used in various algorithms to improve computational efficiency. The Swin transformer also uses a similar idea, dividing the image into multiple nonoverlapping windows, and each window contains $M \times M$ image patches. In this article, M is set to 8. In this way, the self-attention module can be applied to the patches in each window, reducing the number of patches involved in the calculation. Since the feature fusion is only performed in the local area of the image under this kind of window partitioning scheme, and the pixel permutation of the image remains basically unchanged during the whole process, it is unnecessary to add a global positional vector for each patch like ViT. However, considering that the self-attention module ignores the position information of the patches participating in the calculation, the relative position information of patches still needs to be introduced inside the window. For this reason, the Swin transformer adds a relative position bias in the self-attention module, as shown in the following equation:

$$\text{Attention}(Q, K, V) = \text{SoftMax}\left(\frac{QK^T}{\sqrt{d}} + B\right)V \tag{3}$$

where $B \in \mathbb{R}^{M^2\times M^2}$ is the relative position bias, denoting the relative position information between patches within a single window.

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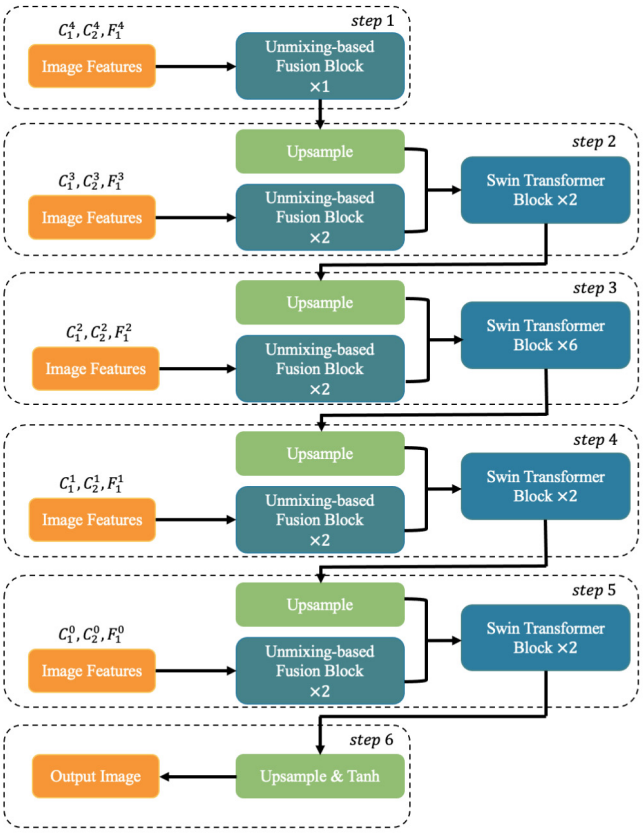


Fig. 4. Architecture of the MFM that is composed of six steps to achieve. The first five steps take the image features generated by FEM (as shown in Fig. 3) as input and complete the fusion of features at different levels. The last step is to upsample the feature map and output the predicted image.

MSA based on the window partitioning scheme in the Swin transformer can greatly reduce the computational complexity compared with the MSA in ViT, but it lacks information across windows, which limits the feature expression ability of patches. Therefore, the FEM in this article adopts the shifted window scheme in the Swin transformer [as shown in Fig. 3(b)], achieving cross-window fusion of patch features through two consecutive self-attention blocks based on different window schemes [W-MSA and SW-MSA, as shown in Fig. 3(c)]. Since these continuous Swin transformer blocks can only bring linear growth to the receptive field, the patch merging layer is also introduced to merge adjacent patches, thereby reducing the image resolution and greatly increasing the receptive field. In this way, combining Swin transformer blocks based on these two kinds of window partitioning schemes, plus a patch merging layer, can form the basic unit of the FEM-Swin extraction block. It can be seen from Fig. 3(a) that the feature extraction network consists of a patch embedding layer (downsampling and linear embedding for input images) and four Swin extraction blocks, and each block performs feature enhancement and double downsampling on the input image features. At the same time, the feature map output by each block will be saved and used in the subsequent MFM. In order to capture the nonlinear difference between coarse images at different dates, this article uses two weight-sharing FEMs to extract coarse features. For the input fine image at the reference date, an FEM with the same structure but different parameters is used to extract fine texture information.

B. Multilevel Fusion Module

In this module, the coarse and fine features obtained in FEM can be fused for fine image generation. The framework of the MFM is shown in Fig. 4, which consists of 6 steps to achieve in total.

First, the features will be fused in an unmixing-based fusion block. According to spectral linear mixing theory, a coarse pixel is composed of multiple fine pixels. If fine pixels can be divided into multiple classes, then the coarse pixel at (x_i, y_i) can be expressed in the following form:

$$C(x_i, y_i) = \sum_{c=0}^{l} f_c(x_i, y_i) F(c) \tag{4}$$

where C and l are the coarse image and the number of fine pixel classes, respectively; f_c is the fraction of fine pixels belong to class c in the coarse pixel; and $F(c)$ is the value of fine pixels that belong to class c. Then, when no land cover type changes between t_1 and t_2, the time change of coarse pixels can be expressed as the weighted sum of the time change of fine pixels that belong to different classes

$$\Delta C(x_i, y_i) = \sum_{c=0}^{l} f_c(x_i, y_i) \Delta F(c). \tag{5}$$

In this way, for a local area containing n coarse pixels, the relationship between ΔC and $\Delta F(c)$ can be expressed by a system of linear mixture equations

$$\begin{bmatrix} \Delta C(x_1, y_1) \\ \vdots \\ \Delta C(x_i, y_i) \\ \vdots \\ \Delta C(x_n, y_n) \end{bmatrix} = \Lambda \begin{bmatrix} \Delta F(1) \\ \vdots \\ \Delta F(c) \\ \vdots \\ \Delta F(l) \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} f_1(x_1, y_1) & f_2(x_1, y_1) & \cdots & f_l(x_1, y_1) \\ \vdots & \vdots & \ddots & \vdots \\ f_1(x_i, y_i) & f_2(x_i, y_i) & \cdots & f_l(x_i, y_i) \\ \vdots & \vdots & \ddots & \vdots \\ f_1(x_n, y_n) & f_2(x_n, y_n) & \cdots & f_l(x_n, y_n) \end{bmatrix}. \tag{6}$$

$\Delta F(c), c = 1, \ldots, l$, could be obtained by computing a least-squares best fit solution based on the inversion of (6). Then, without considering the within-class variability, the time change of each class can be allocated to fine pixels at time t_1 based on (7) to obtain the fine image at time t_2

$$F_2(x_{ij}, y_{ij}) = F_1(x_{ij}, y_{ij}) + \Delta F(c), (x_{ij}, y_{ij}) \in \text{class } c \tag{7}$$

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$$\Lambda' \begin{bmatrix} \Delta C'(x_1, y_1) \\ \vdots \\ \Delta C'(x_i, y_i) \\ \vdots \\ \Delta C'(x_n, y_n) \end{bmatrix} = \begin{bmatrix} \Delta F'(x_1, y_1) \\ \vdots \\ \Delta F'(x_i, y_i) \\ \vdots \\ \Delta F'(x_n, y_n) \end{bmatrix}$$

$$\Lambda' = \begin{bmatrix} w_1(x_1, y_1) & w_1(x_2, y_2) & \cdots & w_1(x_n, y_n) \\ \vdots & \vdots & \ddots & \vdots \\ w_i(x_1, y_1) & w_i(x_2, y_2) & \cdots & w_i(x_n, y_n) \\ \vdots & \vdots & \ddots & \vdots \\ w_n(x_1, y_1) & w_n(x_2, y_2) & \cdots & w_n(x_n, y_n) \end{bmatrix}. \tag{8}$$

In our experiments, the coarse images and the fine images are of the same size, so each coarse pixel can be regarded as a nonlinear weighted sum of surrounding fine pixels. Because the coarse features and the fine features are the nonlinear mappings of the input images, the coarse pixels can be expressed by the linear weighted sum of fine pixels. In addition, since the fine pixels are not classified in advance, this article chooses to treat each fine pixel as an independent individual, so that, for an image area containing n pixels, the inversion of (6) can be changed to (8).

In (8), $\Delta F'$ and $\Delta C'$ denote the temporal changes of fine features and coarse features, respectively. It can be seen from this equation that the time change of each fine pixel can be expressed as the weighted sum of the coarse feature change in the region, so, if the weight $w_i(x_j, y_j)$ is known, $\Delta F'$ can be solved based on weight matrix and $\Delta C'$. To take advantage of data-driven algorithms, this article decides to learn the weight matrix based on the self-attention mechanism and turns (8) into a learnable process.

The core part of the self-attention mechanism is the calculation of matrices query, key, and value, where query and key are used to generate the adjacency matrix between "tokens" in the sequence, and value is used to perform weighted sum of "tokens" based on the adjacency matrix. Then, based on (8), in the unmixing-based fusion block, query and key should be used for generating $w_i(x_j, y_j)$ in the weight matrix to fuse $\Delta C'$. If we use the same input to generate three matrices like the self-attention mechanism, that is to say, using the time change of coarse features to generate query and key, it may allow pixels with the same change but of different classes to be fused together, causing bad effects. Taking the area containing woodland and farmland as an example, under the condition without land-cover change, the time changes in farmland and woodland should be similar. If the self-attention mechanism is used to achieve spatiotemporal fusion, the correlation between woodland features and farmland features will be very high, which will introduce noise to the generation of fine images. Therefore, this article chooses the coarse features C_1' and the fine features F_1' at t_1 to generate query and key, respectively. In this way, when the distance between $C_1'(x_i, y_i)$ and $F_1'(x_j, y_j)$ is small, it means that the features at the two positions are of the same class, which will make the $w_i(x_j, y_j)$ in (8) bigger. When the distance between $C_1'(x_i, y_i)$ and $F_1'(x_j, y_j)$ is large, it means that the mixing fraction of $F_1'(x_j, y_j)$ in $C_1'(x_i, y_i)$ is low, which makes $w_i(x_j, y_j)$ smaller and lets $\Delta C'(x_i, y_i)$ contribute less to $\Delta F'(x_j, y_j)$. However, the above inference is based on the assumption that the mixing fraction of fine pixels in coarse pixels has not changed at two dates, but this situation is often false when land-cover change exists. Therefore, this article introduces a correction factor to adjust the weight matrix, so the weight $w_i(x_j, y_j)$ in (8) could be expressed as

$$w_i(x_j, y_j) = \alpha_i(x_j, y_j) wc_i(x_j, y_j) \tag{9}$$

where $\alpha_i(x_j, y_j)$ is the correction factor and $wc_i(x_j, y_j)$ is the weight calculated by $C_1'(x_i, y_i)$ and $F_1'(x_j, y_j)$. Considering that land-cover change can be reflected in the time change, it is possible to use coarse images at different dates to learn the changes in the mixing fractions. Similar to calculating $wc_i(x_j, y_j)$, this article uses C_1' and C_1' to learn the correction factor α based on the self-attention mechanism. Through the multihead unmixing-based attention (MUA) module proposed above, (1) and (3) can be changed into the following form:

$$\begin{aligned} &Q_1 = C_1' W_{Q_1}, K_1 = F_1' W_{K_1} \\ &Q_2 = C_1' W_{Q_2}, K_2 = C_2' W_{K_2}, V = \Delta C' W_V \\ &\text{Attn} = \text{SoftMax}\left(\frac{Q_1 K_1^T}{\sqrt{d}} + B\right)\text{SoftMax}\left(\frac{Q_2 K_2^T}{\sqrt{d}} + B\right) \\ &\text{Attention}(Q, K, V) \\ &\quad = \text{Norm}(\text{Attn}) V \end{aligned} \tag{10}$$

where Norm(·) means to normalize each row so that the sum of each row is 1.

Like FEM, two consecutive unmixing-based fusion blocks [as shown in Fig. 3(c)] also use different window scheme to achieve cross-window feature fusion. Two consecutive blocks are computed as

$$\begin{aligned} \hat{f}^l &= W - \text{MUA}(\text{LayerNorm}(f^{l-1})) + f^{l-1} \\ f^l &= \text{MLP}(\text{LayerNorm}(\hat{f}^l)) + \hat{f}^l \\ \hat{f}^{l+1} &= \text{SW-MUA}(\text{LayerNorm}(f^l)) + f^l \\ f^{l+1} &= \text{MLP}(\text{LayerNorm}(\hat{f}^{l+1})) + \hat{f}^{l+1} \end{aligned} \tag{11}$$

where $\hat{f}^l$ and f^l denote the output features of the (S)W-MUA module and the MLP module for block l, respectively. Then, the time change of fine features ΔF can be obtained using multiple unmixing-based fusion blocks.

As shown in Fig. 4, after the unmixing-based feature fusion in the first step, MFM will complete multilevel feature fusion in the second to fifth steps. In these steps, the coarse features C_2' used for adding spatial information, the fine features obtained by unmixing-based fusion blocks, and the fine features generated in the previous step will be concatenated and sent to the multiple Swin transformer blocks to obtain the fine features containing multilevel information. Finally, the fine features will go through an Upsample module composed of a pixel shuffle layer, a convolutional layer, and an activation function layer to output the final predicted fine image.

C. Loss Function

This article uses two loss functions to supervise the training process of the proposed model, namely, pixel loss and structure

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loss. Among them, the Charbonnier loss in LapSRN [42] is used for pixel-level image supervision, and the multiscale structural similarity (MS-SSIM) [43] is used to measure the overall similarity between the generated image and the real fine image. These two loss functions are shown in the following equation:

$$L_{\text{structure}} = 1 - \min\left(\text{MS-SSIM}\left(F_2, \hat{F}_2\right) + \varepsilon_s, 1\right)$$
$$L_{\text{pixel}} = \frac{1}{N}\sum_{i=1}^{N}\sqrt{\left(F_2 - \hat{F}_2\right)^2 + {\varepsilon_p}^2} \tag{12}$$

where F_2 and $\hat{F}_2$ are the real fine image and the predicted image at t_2, respectively, and ε_p is used to stabilize the error backpropagation. Since the value of SSIM is hard to reach 1, for training samples that have reached a high SSIM, there is no need to update the network parameters based on the structure loss. Therefore, this article uses the parameter ε_s to reduce the impact of samples with a lower structural loss on the network training process. The values of ε_p and ε_s are set to 0.001 and 0.05, respectively. Finally, the total loss of the network can be expressed by the following equation:

$$L_{\text{total}} = L_{\text{pixel}} + \lambda L_{\text{structure}} \tag{13}$$

where λ is the weight coefficient and the value is set to 1.

III. Experiments and Results

A. Study Areas and Datasets

In the experiment, this article uses public datasets [44] from two locations—Coleambally irrigation area (CIA) and lower Gwydir catchment (LGC)—which represent phenological change and land cover change, respectively. The study area of the CIA is in southern New South Wales. The main vegetation in this area is rice plants, and it has a modern irrigation system. The CIA dataset contains 17 pairs of cloud-free MODIS-Landsat data pairs from 2001 to 2002, each of which has a size of $6 \times 2040 \times 1720$. During the time when the dataset was collected, there was basically no land cover change but the time difference in the CIA area, which can be used to test the performance of SwinSTFM in phenological change prediction. The LGC study area is located in the northern part of New South Wales and contains 14 cloudless data pairs of $6 \times 2720 \times 3200$ from 2004 to 2005. In mid-December 2004, a flood occurred in this area and caused a significant land cover change, which makes the LGC dataset very suitable for testing the performance of SwinSTFM in land cover change prediction. Since the collection areas of these two datasets are mainly covered by plains and farmlands, this article also adds experiments on the AHB dataset [45]. The images of the AHB dataset were collected from Ar Horqin Banner in northeastern China, where the average elevation is higher than that of the CIA and LGC sites. The major industries of Ar Horqin Banner are agriculture and animal husbandry, for which there are a lot of circular pastures and farmlands. The AHB dataset contains 27 cloud-free Landsat-MODIS image pairs with a timespan of more than five years and significant phenological changes.

B. Experiment Design and Evaluation

The overall experiment can be divided into four parts: the results for the LGC dataset, the results for the CIA dataset, the results for the AHB dataset, and the ablation study. In the first three parts, four traditional spatiotemporal fusion methods (STARFM [13], ESTARFM [14], FSDAF [19], and Fit-FC [20]) and two deep learning-based methods (GANSTFM [31] and MSNet [41]) are used for comparison. All three datasets are divided into two parts: training set and test set. In the testing phase, the image pair on November 26, 2004, and the coarse image on December 12, 2004, are used to predict the fine image on December 12, 2004, in the LGC dataset. For the CIA dataset, the image pair on November 25, 2001, and the coarse image on January 12, 2002, are used to predict the fine image on January 12, 2002. For the AHB dataset, the image pair on January 31, 2016, and the coarse image on June 7, 2016, are used to predict the fine image on June 7, 2016. Since ESTARFM and MSNet require two pairs of reference images, this article additionally uses the image pairs on December 28, 2004, February 22, 2002, and October 13, 2016, as inputs. This makes the prediction task much more difficult for algorithms that only need one image pair as for reference images. Specifically, this article only uses the first four channels of the images in the AHB dataset because of the significant strip noise, while all six bands are used in the other two datasets.

For a fair comparison, traditional algorithms and deep learning-based algorithms have adopted similar experimental settings. For the four traditional algorithms, the size of the sliding window for searching similar pixels is uniformly set to 41×41, and the number of similar pixels is 20. At the same time, almost the same dataset generation method and data augment are applied to the deep learning-based algorithms. The three algorithms will randomly select a pair of images at other dates as reference images when generating the training set, and the three input images (MODIS-Landsat images at t_1 and a MODIS image at t_2) will be randomly flipped and rotated during each training iteration. For GANSTFM and SwinSTFM, the input images are 256×256 image patches segmented from the original image with a step size of 128. For MSNet, the input images are 1200×1200 image patches segmented from the original image with a step size of 300. In terms of parameter settings, the initial learning rate of SwinSTFM is 1×10^{-3}, and when the accuracy of the test set no longer rises for three rounds, the learning rate will drop by half. The parameter settings of GANSTFM and MSNet follow the original design.

The performance of the models is evaluated based on six evaluation indices: root mean square error (RMSE), SSIM [46], universal image quality index (UIQI) [47], correlation coefficient (CC), spectral angle mapper (SAM) [48], and erreur relative global adimensionnelle de synthèse (ERGAS) [49]. The ideal values of RMSE, SSIM, UIQI, and CC are 0, 1, 1, and 1, respectively. The closer the values of SAM and ERGAS are to zero, the lower the uncertainty of the fusion result. In addition, in terms of image visualization, this article selects the NIR–red–green channels as RGB. In addition, the average absolute difference (AAD) maps and

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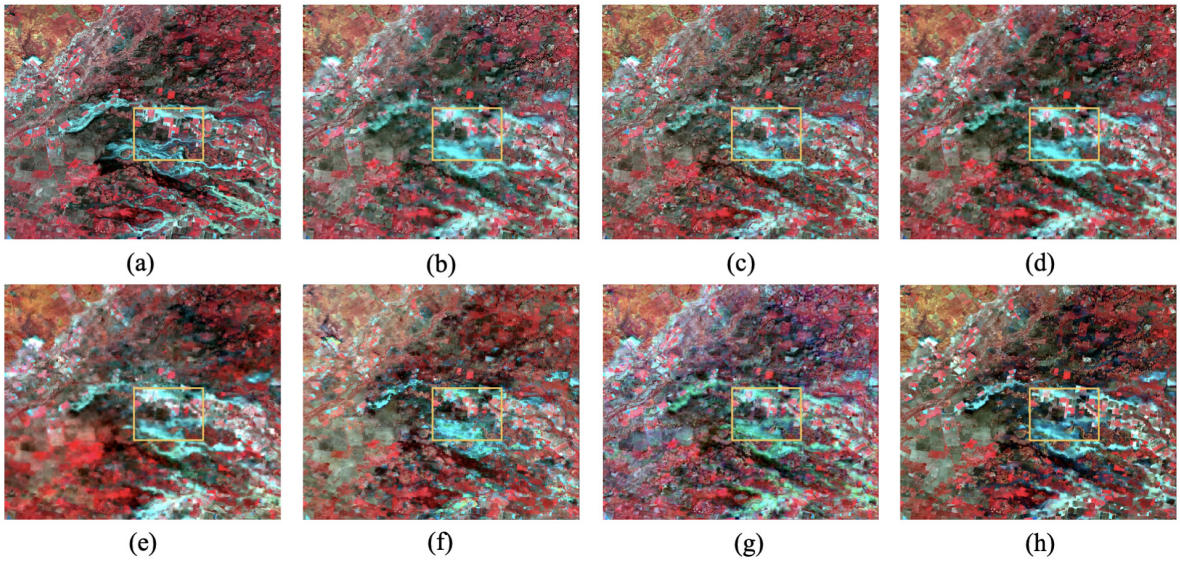


Fig. 5. Fusion results for the LGC dataset on December 12, 2004, with different methods. (a) Real image. (b) STARFM. (c) ESTARFM. (d) FSDAF. (e) Fit-FC. (f) GANSTFM. (g) MSNet. (h) SwinSTFM.

scatter plots are also used to compare algorithm performance. The former reflects the average difference in pixels between normalized predicted images and real images, and the latter describes the number of pixel pairs of the predicted image and the real image in different intervals. More dataset details can be found in the Supplementary Material.

C. Fusion Results for LGC Dataset

Fig. S1 in the Supplementary Material shows three test image pairs and their corresponding subregions. It can be seen that the land-cover types on December 12, which are very different from the images of the other two dates, changed drastically due to the flood. This makes the spatiotemporal fusion task extremely difficult. Besides, compared to November 26, the image pair on December 28 and the image pair on December 12 are more similar in land-cover types and are also nearly identical in some texture details (as can be seen from the subregions in Fig. S1 in the Supplementary Material). Therefore, ESTARFM and MSNet, which use the image pair of December 28 as additional input, are more dominant in spatiotemporal fusion.

The quantitative comparisons of fusion results are listed in Table I. The prediction of BiSwinSTFM is obtained by averaging two images generated from SwinSTFM using November 26 and December 28 as reference dates, respectively. As can be seen from the table, the improved traditional methods are not much better than STARFM under the same settings. At the same time, although the deep learning methods do not have a gap with the traditional methods in the performance of the first four bands, they achieve outstanding performance in the latter two bands. Furthermore, compared with the six methods participating in the comparison, SwinSTFM achieves the best on almost all indices, only slightly inferior to MSNet on one index. If BiSwinSTFM is taken into account, the algorithm proposed in this article will achieve the best on all indices and outperform other algorithms by a large margin. These quantitative results demonstrate that SwinSTFM can be well adapted to scenes with drastic land-cover change.

Figs. 5 and 6 show the predicted images and subregions of the seven spatiotemporal fusion algorithms on December 12. As can be seen from the figures, the fusion results of SwinSTFM have the highest degree of restoration in appearance and land-cover type. In addition, it can be found that, although the results of MSNet are very similar to the original image in land cover, they have a large difference in appearance, which is affected by the image pair on December 28. This also proves the robustness of the algorithm proposed in this article. Fig. 7 shows the comparison on the AAD maps of the fusion results in subregions. It can be seen that SwinSTFM achieves the best fusion effect on the pixels whose land-cover type is water, while there are only very few pixels containing water in the reference images. This shows that SwinSTFM has strong generalizability and learning ability from a few samples. Fig. 8 shows the scatter plots of predicted image values and real image values. The narrower the scatter area is, the higher the similarity of pixels will be. It can be seen intuitively that the generated image of SwinSTFM is more similar to the real image and has the highest coefficient of determination. From all the above results, it can be found that SwinSTFM has excellent performance on spatiotemporal fusion tasks with land-cover change. An important application of spatiotemporal fusion is obtaining high-quality remote sensing image time series to generate NDVI data, which is also the purpose of this article. Therefore, the calculated NDVI based on the predicted fine images is presented in Fig. 9. The NDVI series generated based on the fused remote sensing images can be used for further applications, such as ecosystem monitoring.

D. Fusion Results for CIA Dataset

Fig. S2 in the Supplementary Material shows three test image pairs and their corresponding subregions. Due to the long time span, the phenological differences between

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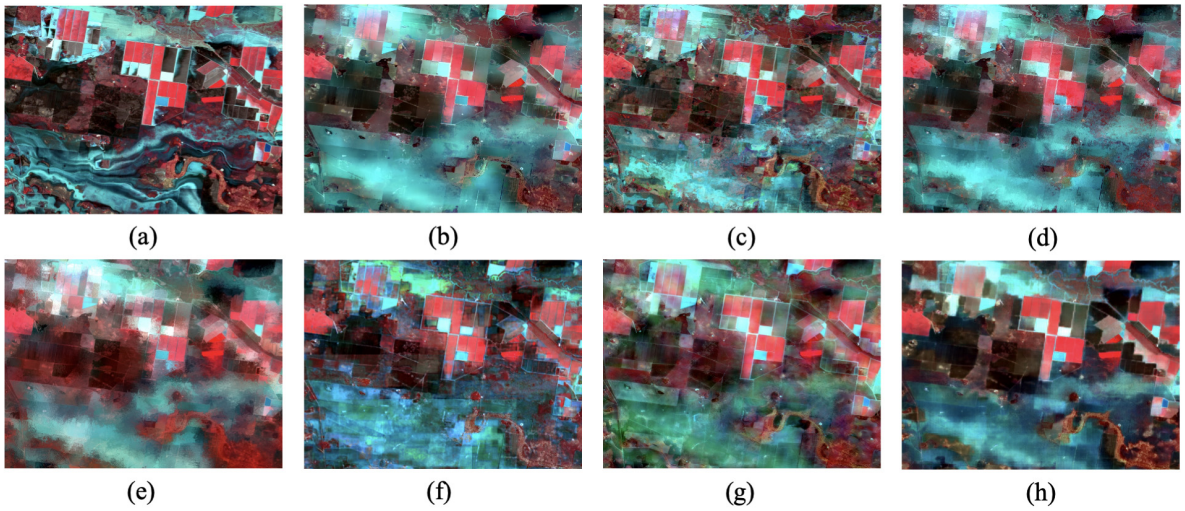


Fig. 6. Fusion results for the subregions on December 12, 2004, with different methods. (a) Ground truth. (b) STARFM. (c) ESTARFM. (d) FSDAF. (e) Fit-FC. (f) GANSTFM. (g) MSNet. (h) SwinSTFM.

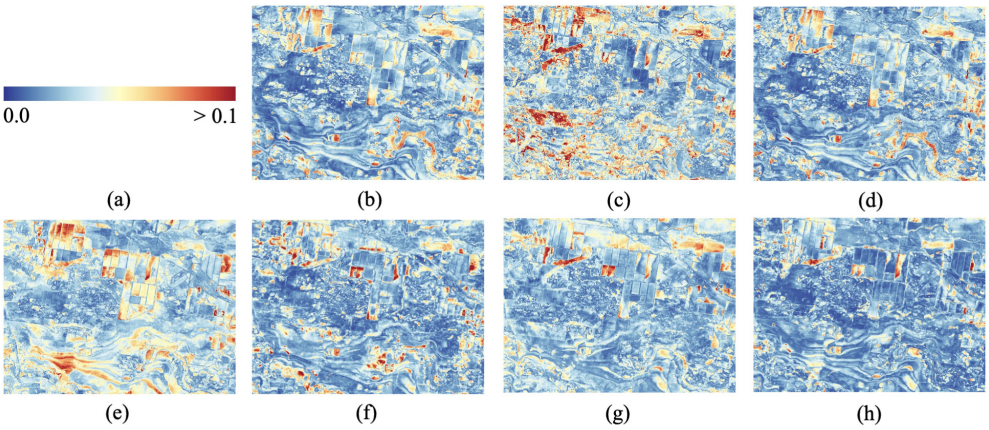


Fig. 7. AAD maps for the subregions on December 12, 2004, with different methods. (a) Color bar. (b) STARFM. (c) ESTARFM. (d) FSDAF. (e) Fit-FC. (f) GANSTFM. (g) MSNet. (h) SwinSTFM.

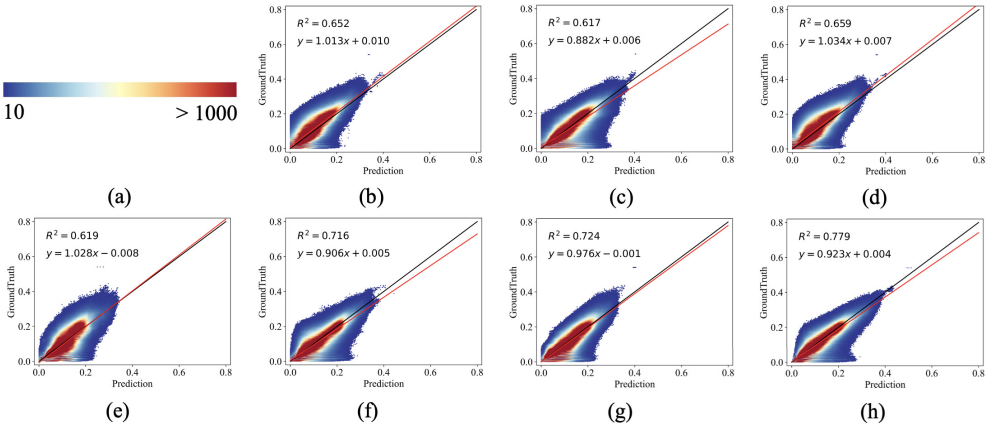


Fig. 8. Scatter plots for the fusion results on December 12, 2004, with different methods. (a) Color bar. (b) STARFM. (c) ESTARFM. (d) FSDAF. (e) Fit-FC. (f) GANSTFM. (g) MSNet. (h) SwinSTFM.

the three image pairs are obvious from the appearance. Specifically, the images on November 25 are more similar to the images on January 12 in land-cover types, but there is a larger phenological difference between them. The images on February 22 have a similar appearance to the images on January 12 but with some differences in texture details. Therefore, for the algorithm that only needs to use the image pair on November 25 as the reference input, the biggest difficulty lies in obtaining the time change information.

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TABLE I
QUANTITATIVE EVALUATION RESULTS FOR THE LGC DATASET ON DECEMBER 12, 2004, WITH DIFFERENT METHODS. THE BOLD VALUES INDICATE THE BEST RESULTS

Band	STARFM	ESTARFM	FSDAF	Fit-FC	GANSTFM	MSNet	SwinSTFM	BiSwinSTFM
				RMSE				
Blue	0.0152	0.0148	0.0148	0.0178	0.0153	0.0167	**0.0128**	(**0.0123**)
Green	0.0212	0.0205	0.0205	0.0256	0.0213	0.02	**0.0191**	(**0.0187**)
Red	0.0263	0.0259	0.0255	0.0331	0.0266	0.0243	**0.0238**	(**0.023**)
NIR	0.0393	0.0389	0.0384	0.0469	0.041	0.0373	**0.0328**	(**0.0307**)
SWIR1	0.0615	0.0653	0.0612	0.0618	0.0542	0.0538	**0.0492**	(**0.0452**)
SWIR2	0.0509	0.0559	0.0509	0.0454	0.0396	0.0427	**0.0361**	(**0.0329**)
				SSIM				
Blue	0.9144	0.9158	0.9126	0.907	0.9165	0.9108	**0.9337**	(**0.9391**)
Green	0.8747	0.878	0.8708	0.861	0.8769	0.8925	**0.8954**	(**0.9035**)
Red	0.8391	0.8428	0.8332	0.8137	0.8417	**0.8646**	0.8643	(**0.8748**)
NIR	0.7446	0.7242	0.7364	0.7138	0.7134	0.7531	**0.7764**	(**0.8033**)
SWIR1	0.5483	0.5348	0.5301	0.5077	0.5973	0.5986	**0.6445**	(**0.6858**)
SWIR2	0.5897	0.5498	0.5834	0.5848	0.6672	0.6541	**0.708**	(**0.7449**)
				UIQI				
Blue	0.6972	0.7272	0.7119	0.6421	0.6423	0.6287	**0.7636**	(**0.7766**)
Green	0.6830	0.7105	0.6973	0.6393	0.6626	0.6526	**0.7536**	(**0.7685**)
Red	0.6966	0.7203	0.7041	0.6373	0.6836	0.7155	**0.7617**	(**0.7773**)
NIR	0.7879	0.7825	0.7939	0.6194	0.7524	0.7592	**0.8482**	(**0.8678**)
SWIR1	0.7298	0.6780	0.7287	0.6566	0.7946	0.7628	**0.8349**	(**0.8582**)
SWIR2	0.6723	0.6048	0.6632	0.6494	0.7963	0.7463	**0.8388**	(**0.8622**)
				CC				
Blue	0.6983	0.7299	0.7135	0.6479	0.6496	0.6526	**0.765**	(**0.7788**)
Green	0.6841	0.713	0.6997	0.6462	0.6694	0.6873	**0.7562**	(**0.7716**)
Red	0.6979	0.7233	0.7075	0.6488	0.6874	0.7348	**0.7625**	(**0.7784**)
NIR	0.8065	0.7893	0.8149	0.6725	0.7592	0.8027	**0.8512**	(**0.8706**)
SWIR1	0.7709	0.696	0.7736	0.7095	0.8031	0.7984	**0.8417**	(**0.865**)
SWIR2	0.751	0.6148	0.753	0.7128	0.8039	0.7834	**0.844**	(**0.8674**)
				SAM				
	17.2352	17.754	16.7932	18.3156	16.5068	16.3468	**14.432**	(**13.6468**)
				ERGAS				
	2.2217	2.32	2.1947	2.3381	1.998	2.0143	**1.7731**	(**1.6634**)

Table II shows the quantitative comparisons of fusion results. Since there was no substantial land-cover change in the CIA area during the observation period, the spatiotemporal fusion task becomes much easier, and the gap between traditional methods and deep learning-based methods is greatly reduced. However, SwinSTFM still has remarkable performance and achieves the best results in all indices, which proves the superiority and generalization of the algorithm proposed in this article once again. Two other interesting things can be discovered from the table. The first is that the performance of MSNet on the CIA dataset is slightly worse than that of GANSTFM. This is because the image pair on November 25 has provided enough texture information used to predict the fine image on January 12, and MSNet cannot obtain more useful information from another pair of reference images. Another thing is that BiSwinSTFM outperforms SwinSTFM in only a few indices. This makes sense because BiSwinSTFM just mechanically combines the two generated images at a fixed scale. When one of the predicted images is not significantly better than the other, the fusion result will not get better. It is also interesting that BiSwinSTFM outperforms MSNet on most indices, while the similar fusion process based on two reference dates has been sufficiently learned in the latter algorithm.

Figs. 10 and 11 show the predicted images and subregions of the seven spatiotemporal fusion algorithms on January 12. It can be seen that, compared with the LGC dataset, the performance of the spatiotemporal fusion methods on the CIA dataset has been significantly improved, and there is not much difference between them, indicating that land-cover

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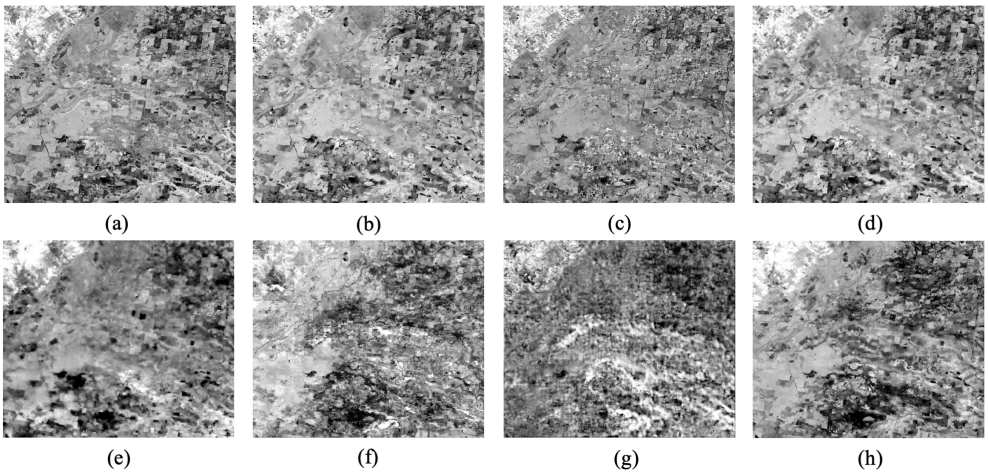


Fig. 9. NDVI of the LGC dataset on December 12, 2004, with different methods. (a) Real image. (b) STARFM. (c) ESTARFM. (d) FSDAF. (e) Fit-FC. (f) GANSTFM. (g) MSNet. (h) SwinSTFM.

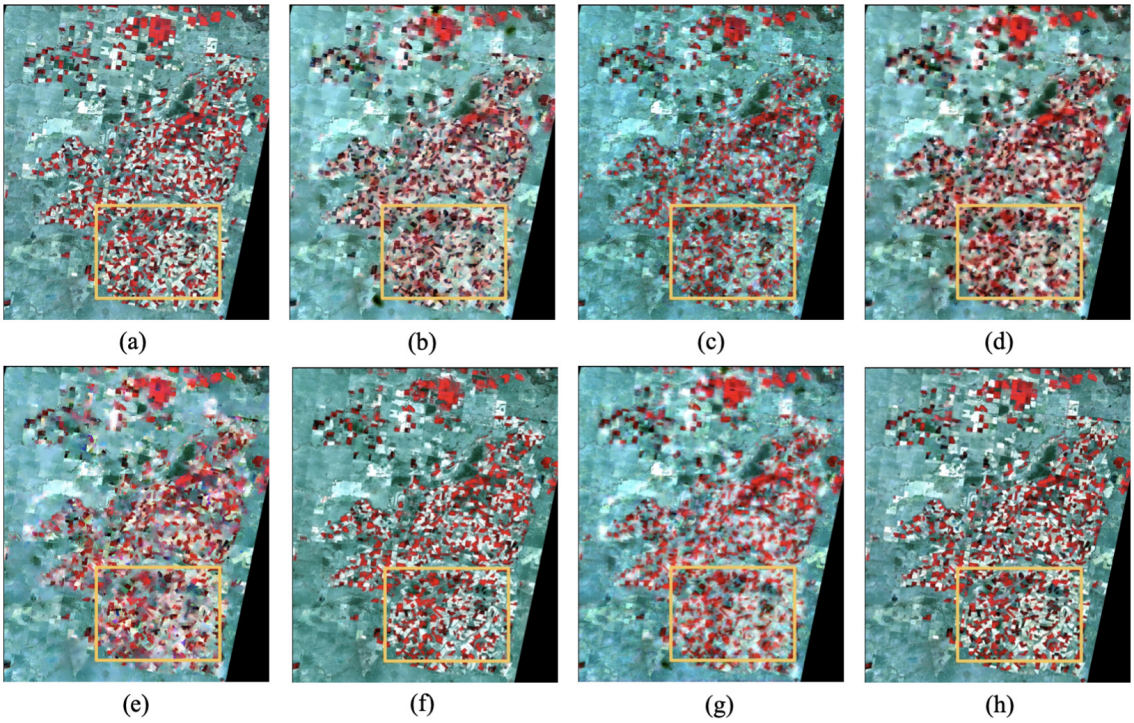


Fig. 10. Fusion results for the CIA dataset on January 12, 2002, with different methods. (a) Real image. (b) STARFM. (c) ESTARFM. (d) FSDAF. (e) Fit-FC. (f) GANSTFM. (g) MSNet. (h) SwinSTFM.

change is still more difficult to learn than the phenological change in spatiotemporal fusion tasks. However, in terms of specific texture details, GANSTFM and SwinSTFM are significantly better than other methods, which can also be seen from Table II. At the same time, SwinSTFM is superior in appearance and texture (as can be seen in Fig. 11) to above two methods. These characteristics are evident in the AAD maps shown in Fig. 12. The fusion methods except GANSTFM and SwinSTFM have a lot of errors in texture details, and GANSTFM is slightly inferior to SwinSTFM. Fig. 13 shows the scatter plots of predicted image values and real image values. Since phenological change is easier to predict, all the spatiotemporal fusion methods involved in the comparison have higher coefficients of determination, and the gap between traditional methods and deep learning-based methods is greatly reduced. However, SwinSTFM still maintains a big advantage over other methods. It can be seen from the above results that SwinSTFM still has an excellent performance in spatiotemporal fusion with phenological change, which fully proves the universality and superiority of SwinSTFM in spatiotemporal fusion tasks.

E. Fusion Results for AHB Dataset

Fig. S3 in the Supplementary Material shows three test image pairs and their corresponding subregions. Time spans of up to nine months allow for large phenological and

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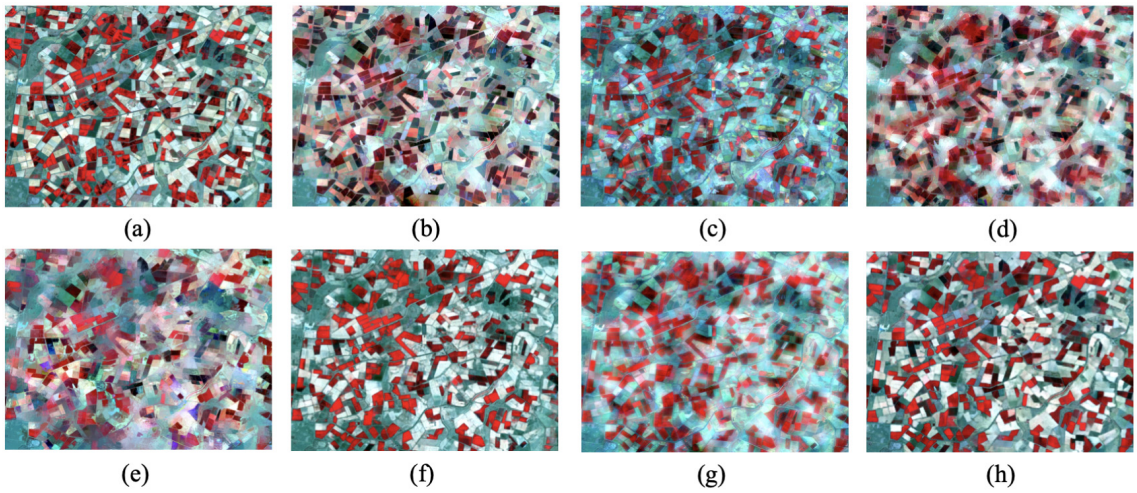


Fig. 11. Fusion results for the subregions on January 12, 2002, with different methods. (a) Ground truth. (b) STARFM. (c) ESTARFM. (d) FSDAF. (e) Fit-FC. (f) GANSTFM. (g) MSNet. (h) SwinSTFM.

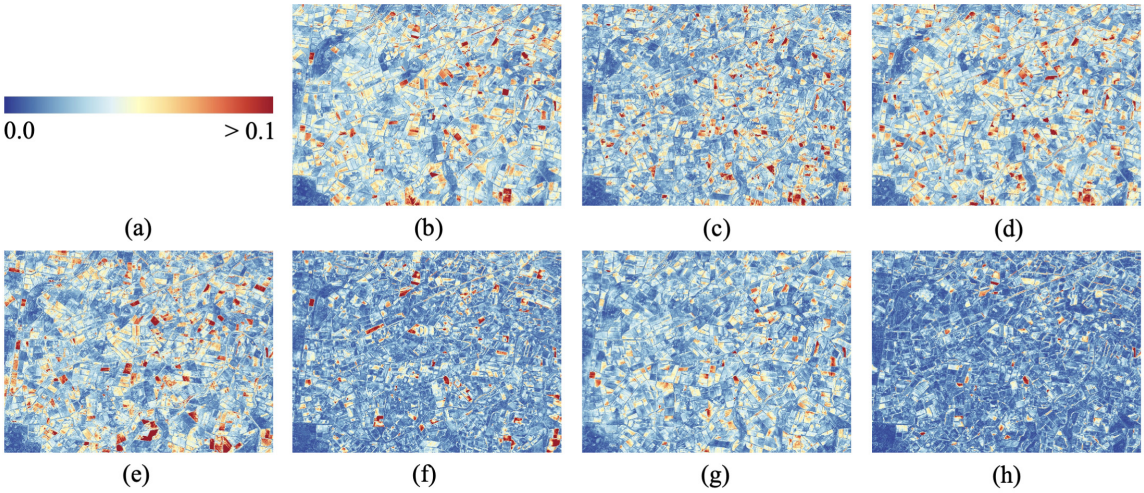


Fig. 12. AAD maps for the subregions on January 12, 2002, with different methods. (a) Color bar. (b) STARFM. (c) ESTARFM. (d) FSDAF. (e) Fit-FC. (f) GANSTFM. (g) MSNet. (h) SwinSTFM.

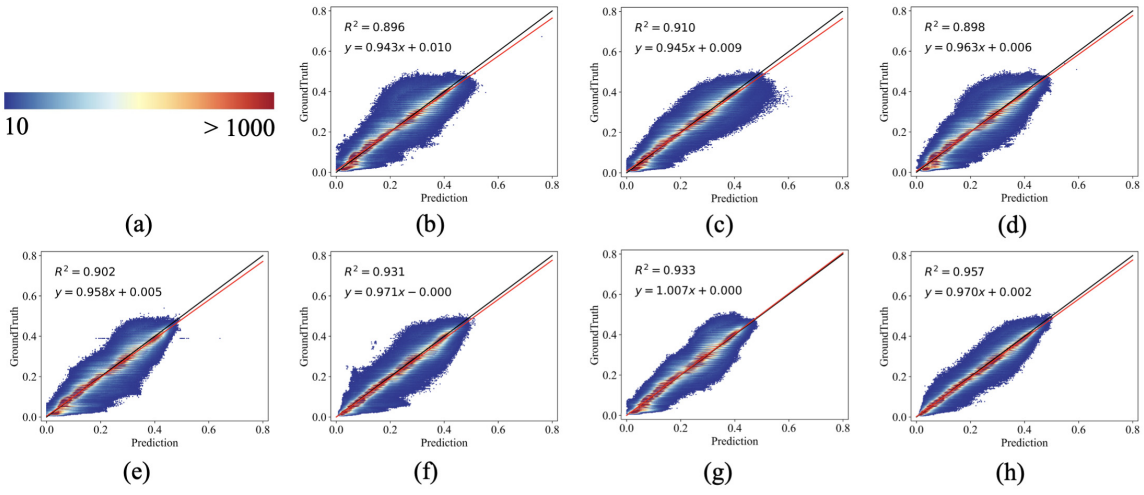


Fig. 13. Scatter plots for the fusion results on January 12, 2002, with different methods. (a) Color bar. (b) STARFM. (c) ESTARFM. (d) FSDAF. (e) Fit-FC. (f) GANSTFM. (g) MSNet. (h) SwinSTFM.

land-cover changes between the three image pairs. In addition, in experiments, we find that the performance of GANSTFM drops rapidly after adding the autoencoder module. Therefore, in the AHB dataset, we replace all the autoencoder features

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TABLE II
QUANTITATIVE EVALUATION RESULTS FOR THE CIA DATASET ON JANUARY 12, 2002, WITH DIFFERENT METHODS. THE BOLD VALUES INDICATE THE BEST RESULTS

Band	STARFM	ESTARFM	FSDAF	Fit-FC	GANSTFM	MSNet	SwinSTFM	BiSwinSTFM
				RMSE				
Blue	0.0158	0.0126	0.0158	0.016	0.0147	0.0125	**0.0103**	(0.0109)
Green	0.0234	0.0198	0.0239	0.0241	0.0187	0.0199	**0.0143**	(0.0152)
Red	0.0366	0.0352	0.0393	0.038	0.0291	0.033	**0.0226**	(0.023)
NIR	0.0551	0.0541	0.0514	0.0437	0.0403	0.0437	**0.0324**	(0.0342)
SWIR1	0.0468	0.0413	0.0459	0.0497	0.0397	0.035	**0.0316**	(0.0386)
SWIR2	0.0378	0.0337	0.0381	0.04	0.0351	0.0299	**0.0282**	(0.0318)
				SSIM				
Blue	0.8986	0.924	0.8895	0.8887	0.9225	0.929	**0.9481**	(0.9391)
Green	0.8589	0.8836	0.8384	0.8394	0.9015	0.8894	**0.9251**	(0.9191)
Red	0.7939	0.7981	0.7452	0.762	0.8516	0.8271	**0.8859**	(**0.8864**)
NIR	0.6768	0.7014	0.6558	0.6895	0.7683	0.74	**0.8165**	(**0.8186**)
SWIR1	0.7594	0.778	0.7204	0.7132	0.8073	0.8296	**0.8468**	(**0.855**)
SWIR2	0.7885	0.8078	0.7569	0.7389	0.8196	0.846	**0.856**	(**0.8651**)
				UIQI				
Blue	0.8275	0.8968	0.8234	0.8262	0.8987	0.8968	**0.9423**	(0.938)
Green	0.8367	0.8882	0.8266	0.8326	0.9161	0.885	**0.9487**	(0.9403)
Red	0.8475	0.8681	0.8179	0.8418	0.9178	0.8762	**0.9519**	(0.9483)
NIR	0.811	0.8393	0.8293	0.8738	0.9021	0.871	**0.9357**	(0.9235)
SWIR1	0.9265	0.9409	0.9257	0.9159	0.9464	0.9556	**0.9669**	(0.9491)
SWIR2	0.9262	0.9411	0.9215	0.9121	0.9377	0.9517	**0.961**	(0.9496)
				CC				
Blue	0.847	0.9024	0.8459	0.8376	0.9043	0.9039	**0.9435**	(0.9416)
Green	0.8561	0.8968	0.8499	0.8433	0.9177	0.8986	**0.949**	(0.9434)
Red	0.8681	0.8775	0.8462	0.8538	0.9186	0.8993	**0.9519**	(0.9498)
NIR	0.8117	0.8415	0.8319	0.879	0.9022	0.8789	**0.9362**	(0.9299)
SWIR1	0.9271	0.941	0.9261	0.9182	0.9467	0.9563	**0.9673**	(0.9508)
SWIR2	0.9264	0.9412	0.922	0.9142	0.9385	0.9521	**0.9617**	(0.951)
				SAM				
	11.0121	9.8456	11.1193	11.0077	8.8099	9.0446	**6.939**	(7.4703)
				ERGAS				
	1.2853	1.148	1.2987	1.2865	1.0788	1.0603	**0.8229**	(0.8884)

in the GANSTFM generator loss with images, denoted by GANSTFM*.

Table III shows the quantitative comparisons of fusion results. Due to the large difference between image pairs in the AHB dataset, the performance of all spatiotemporal fusion algorithms has a significant drop compared to the first two datasets. However, the deep learning-based methods show excellent performance far exceeding the traditional methods, which also reflects the superiority of deep learning models in difficult spatiotemporal fusion tasks. Furthermore, SwinSTFM still achieves the relatively best results, which again proves the generality of the proposed algorithm in this article.

Figs. 14 and 15 show the predicted images and subregions of the seven spatiotemporal fusion algorithms on June 7. Overall, GANSTFM and SwinSTFM can restore the fine image on June 7 well, while other methods get very bad results. In terms of details, SwinSTFM performs better than GANSTFM, and the transition in detail is more natural. These characteristics are more evident in Figs. 16 and 17.

F. Ablation Study

In this section, this article designs three models to reflect the advantages of SwinSTFM for the problems of deep learning-based methods. The three models are SwinSTFM without multilevel, XcepSTFM, and SwinSTFM without unmixing. The first model still uses the Swin transformer as the backbone, but no downsampling is performed after the patch embedding layer. The memory usage of the self-attention mechanism will increase sharply as the resolution increases. Therefore, the number of parameters of SwinSTFM without multilevel

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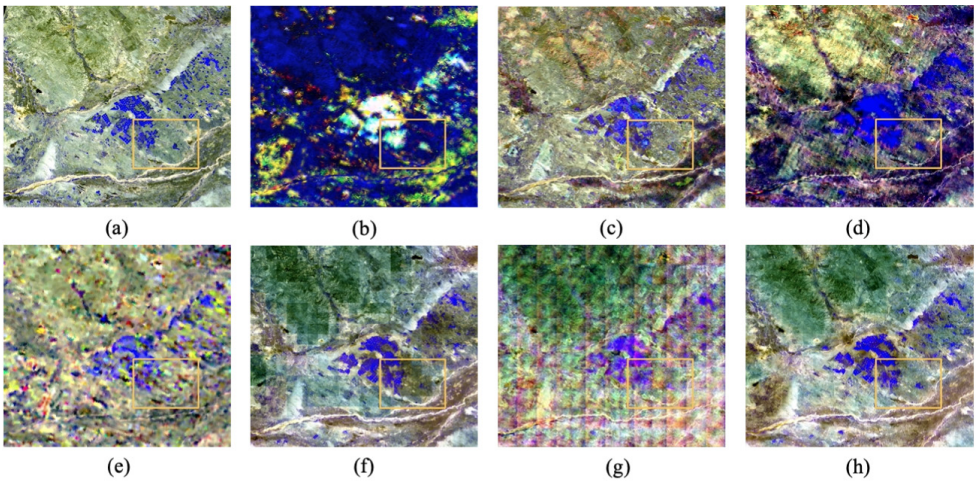


Fig. 14. Fusion results for the AHB dataset on June 7, 2016, with different methods. (a) Real image. (b) STARFM. (c) ESTARFM. (d) FSDAF. (e) Fit-FC. (f) GANSTFM*. (g) MSNet. (h) SwinSTFM.

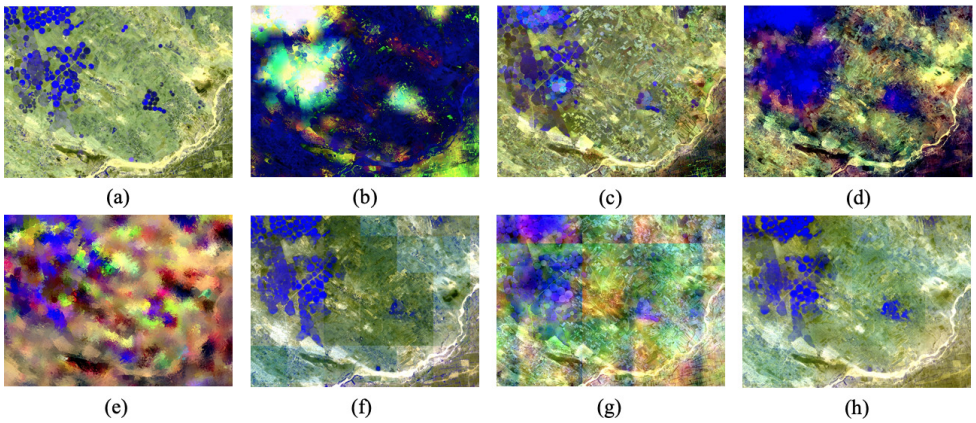


Fig. 15. Fusion results for the subregions on June 7, 2016, with different methods. (a) Ground truth. (b) STARFM. (c) ESTARFM. (d) FSDAF. (e) Fit-FC. (f) GANSTFM*. (g) MSNet. (h) SwinSTFM.

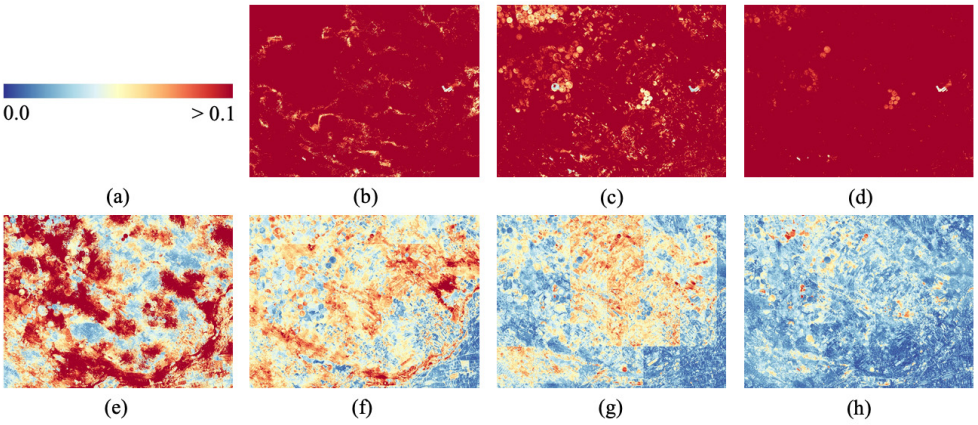


Fig. 16. AAD maps for the subregions on June 7, 2016, with different methods. (a) Color bar. (b) STARFM. (c) ESTARFM. (d) FSDAF. (e) Fit-FC. (f) GANSTFM*. (g) MSNet. (h) SwinSTFM.

is greatly reduced, and it is only one-fifth of SwinSTFM in the case of occupying similar video memory. XcepSTFM replaces all transformer-related modules in SwinSTFM with Xception [50] to compare the performance of CNN and transformer in spatiotemporal fusion under the same conditions. The third model is to replace the unmixing-based fusion blocks in MFM with the direct addition of fine features and time changes of coarse features to see the impact of the unmixing module on image fusion. Table IV shows the comparison of the parameters of three models with SwinSTFM. It can be seen that the parameters of the latter two models used for comparison and SwinSTFM are in the same order of magnitude, thus

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TABLE III

QUANTITATIVE RESULTS FOR THE AHB DATASET ON JUNE 7, 2016, WITH DIFFERENT METHODS. THE BOLD VALUES INDICATE THE BEST RESULTS

Band	STARFM	ESTARFM	FSDAF	Fit-FC	GANSTFM*	MSNet	SwinSTFM	BiSwinSTFM
				RMSE				
Blue	0.3336	0.1106	0.1085	0.0798	0.0599	**0.0462**	0.0540	(0.0618)
Green	0.2497	0.1047	0.1162	0.0577	0.0494	0.0647	**0.0433**	(0.0506)
Red	0.3012	0.1174	0.1393	0.0795	0.0488	0.0818	**0.0417**	(0.0474)
NIR	0.1917	0.1606	0.1893	0.1029	0.0525	0.0959	**0.0432**	(0.0487)
				SSIM				
Blue	0.1481	0.3421	0.2419	0.5980	0.7916	**0.8539**	0.8183	(0.7921)
Green	0.2083	0.4618	0.2852	0.6975	0.8175	0.7604	**0.8344**	(0.8296)
Red	0.1690	0.4939	0.2811	0.5339	0.7763	0.6914	**0.7809**	(**0.8017**)
NIR	0.4630	0.5614	0.4718	0.5240	0.7773	0.7147	**0.7790**	(**0.7814**)
				UIQI				
Blue	-0.0433	0.1513	0.0541	**0.3306**	0.1600	-0.0211	0.1852	(0.2181)
Green	-0.0760	0.2918	0.1177	0.3515	0.4599	0.1762	**0.4961**	(**0.5303**)
Red	-0.1374	0.3257	0.1319	0.4054	0.6337	0.3239	**0.6706**	(**0.7002**)
NIR	0.1675	0.4765	0.2697	0.3581	0.6366	0.3662	**0.6512**	(**0.6571**)
				CC				
Blue	-0.4152	0.4336	**0.6401**	0.4481	0.1864	-0.0231	0.2167	(0.2564)
Green	-0.3499	0.6108	**0.5816**	0.3815	0.4865	0.2025	0.5146	(0.5687)
Red	-0.4796	0.5469	0.6366	0.4360	0.6557	0.4596	**0.6851**	(**0.7351**)
NIR	0.2718	**0.6584**	0.4655	0.4118	0.6426	0.4247	0.6570	(**0.6704**)
				SAM				
	47.8834	19.4082	25.9913	21.3767	12.4235	13.5187	**11.9162**	(**11.2326**)
				ERGAS				
	10.5609	4.2094	5.2817	2.8123	1.9479	2.4136	**1.7124**	(1.9677)

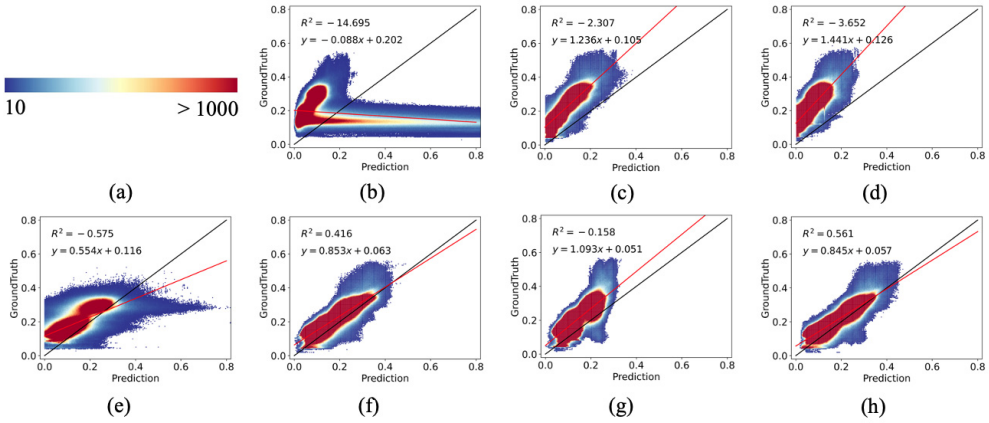


Fig. 17. Scatter plots for the fusion results on June 7, 2016, with different methods. (a) Color bar. (b) STARFM. (c) ESTARFM. (d) FSDAF. (e) Fit-FC. (f) GANSTFM*. (g) MSNet. (h) SwinSTFM.

TABLE IV

PARAMETERS' COMPARISON FOR ABLATION STUDY

Model	SwinSTFM w/o multi-level	XcepSTFM	SwinSTFM w/o unmixing	SwinSTFM
Parameters	784,628	37,923,334	30,719,508	39,665,893

ensuring a fair comparison environment. Table V shows the performance of the four algorithms on the LGC dataset, and SwinSTFM achieves the best results on almost all indices. This article also selects two subregions to study the performance improvement brought by the transformer architecture and unmixing module in SwinSTFM. Fig. 18 shows the related

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TABLE V

QUANTITATIVE EVALUATION RESULTS FOR THE LGC DATASET ON DECEMBER 12, 2004, WITH DIFFERENT METHODS. THE BOLD VALUES INDICATE THE BEST RESULTS

Index	SwinSTFM w/o multi-level	XcepSTFM	SwinSTFM w/o unmixing	SwinSTFM
RMSE	0.0305	0.0313	0.0305	**0.0290**
SSIM	0.7894	0.7854	0.7976	**0.8037**
UIQI	0.7839	0.7752	0.7857	**0.8001**
CC	0.7879	0.7767	0.7868	**0.8034**
SAM	15.8057	15.5547	15.2629	**14.4320**
ERGAS	1.8503	1.8903	1.8685	**1.7731**

TABLE VI

QUANTITATIVE EVALUATION RESULTS FOR THE LGC DATASET ON DECEMBER 12, 2004, WITH DIFFERENT METHODS. THE BOLD VALUES INDICATE THE BEST RESULTS

Index	L1	Charbonnier	L1 + MS-SSIM	Charbonnier + MS-SSIM	L1 + Clipped MS-SSIM	Original
RMSE	0.0291	0.0293	0.0296	0.0296	**0.0290**	**0.0290**
SSIM	0.7966	0.7954	0.8024	0.8020	**0.8044**	0.8037
UIQI	0.7915	0.7872	0.7984	0.7998	**0.8004**	0.8001
CC	0.7972	0.7933	0.8001	0.8026	0.8021	**0.8034**
SAM	14.5645	14.7083	14.8726	14.6252	14.6663	**14.4320**
ERGAS	1.7898	1.8025	1.8239	1.8206	1.7882	**1.7731**

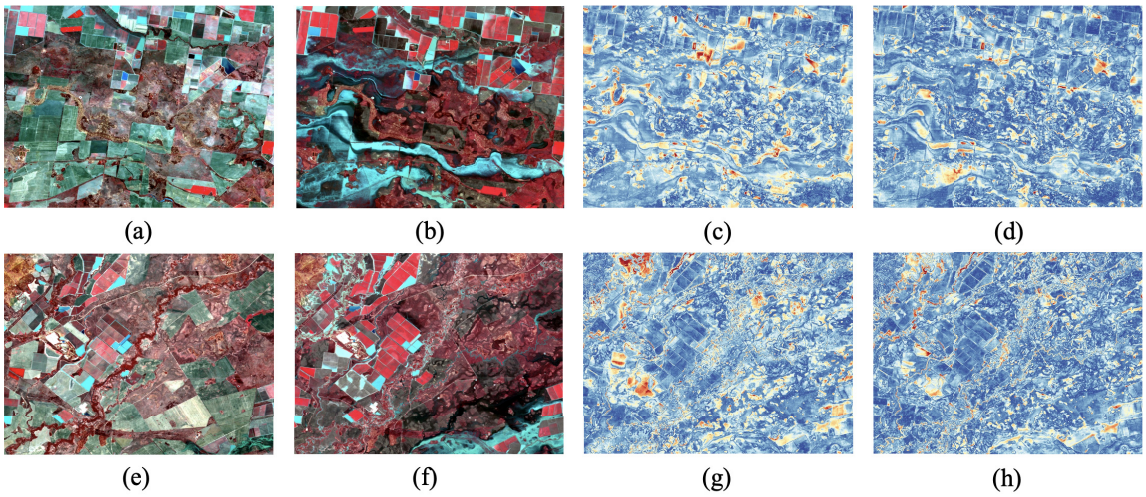


Fig. 18. Subregions and AAD maps on the LGC dataset. (a) Subregion 1 on November 26, 2004. (b) Subregion 1 on December 12, 2004. (c) AAD map of XcepSTFM on subregion 1. (d) AAD map of SwinSTFM on subregion 1. (e) Subregion 2 on November 26, 2004. (f) Subregion 2 on December 12, 2004. (g) AAD map of SwinSTFM w/o unmixing on subregion 2. (h) AAD map of SwinSTFM on subregion 2.

results, and these two modules make SwinSTFM achieve better fusion results on image regions with land-cover changes. These experiments show that SwinSTFM can well solve the three problems raised at the beginning of this article.

Furthermore, this article also conducts five experiments to testify to the loss function, as shown in Table VI. The loss function used in this article consists of two parts: pixel loss and structure loss. The pixel loss uses the Charbonnier loss in LapSRN, and the structure loss uses the clipped MS-SSIM. It can be seen from Table VI that the impact of different loss functions on the performance of the model is very small, which shows that the excellent results obtained on multiple datasets are derived from the network design of SwinSTFM.

IV. CONCLUSION

To address several problems of deep learning-based spatiotemporal fusion methods, this article proposes an algorithm named SwinSTFM that combines advanced deep learning models and linear spectral mixing theory. The algorithm mainly includes two modules: FEM and MFM. FEM completes the multilevel feature generation of the input images, and MFM completes the feature fusion of different levels based on the unmixing theories and self-attention mechanism. Through many experimental results and comparisons with six mainstream methods, the generalization and superiority of SwinSTFM on different types of remote sensing fusion tasks can be proved.

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Although SwinSTFM has achieved considerable performance improvements over current spatiotemporal fusion methods, there is still a lot of room for improvement. First, due to the large number of parameters in SwinSTFM, only using some data augment methods can reduce the influence of overfitting in the experimental part. At the same time, there is no doubt that the performance of SwinSTFM can be further improved by establishing a larger-scale spatiotemporal fusion dataset or using parameters of models pretrained on other image tasks. Second, this article opens a new path that combines excellent deep learning models with traditional fusion methods, which hopefully can inspire more meaningful research. In addition, it is difficult to obtain image pairs without registration errors in real work, and our future work will also focus on how to make the model less sensitive to errors.

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The open-source code of this article can be downloaded on https://github.com/LouisChen0104/swinstfm.

REFERENCES

[1] E. Nduati, Y. Sofue, A. Matniyaz, J. Park, W. Yang, and A. Kondoh, "Cropland mapping using fusion of multi-sensor data in a complex urban/peri-urban area," *Remote Sens.*, vol. 11, no. 2, p. 207, Jan. 2019.

[2] M. Zhang, H. Lin, H. Sun, and Y. Cai, "Estimation of vegetation productivity using a Landsat 8 time series in a heavily urbanized area, central China," *Remote Sens.*, vol. 11, no. 2, p. 133, Jan. 2019.

[3] T. Hwang, C. Song, P. V. Bolstad, and L. E. Band, "Downscaling real-time vegetation dynamics by fusing multi-temporal MODIS and landsat NDVI in topographically complex terrain," *Remote Sens. Environ.*, vol. 115, no. 10, pp. 2499–2512, Oct. 2011.

[4] P. Arévalo, P. Olofsson, and C. E. Woodcock, "Continuous monitoring of land change activities and post-disturbance dynamics from Landsat time series: A test methodology for REDD+ reporting," *Remote Sens. Environ.*, vol. 238, Mar. 2020, Art. no. 111051.

[5] E. Hamunyela, P. Brandt, D. Shirima, H. T. T. Do, M. Herold, and R. M. Roman-Cuesta, "Space-time detection of deforestation, forest degradation and regeneration in montane forests of eastern Tanzania," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 88, Jun. 2020, Art. no. 102063.

[6] R. Interdonato, D. Ienco, R. Gaetano, and K. Ose, "DuPLO: A DUal view point deep learning architecture for time series classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 149, pp. 91–104, Mar. 2019.

[7] H. A. Ghrefat and P. C. Goodell, "Land cover mapping at alkali flat and lake Lucero, white sands, New Mexico, USA using multi-temporal and multi-spectral remote sensing data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 13, no. 4, pp. 616–625, Aug. 2011.

[8] D. Jia, P. Gao, C. Cheng, and S. Ye, "Multiple-feature-driven co-training method for crop mapping based on remote sensing time series imagery," *Int. J. Remote Sens.*, vol. 41, no. 20, pp. 8096–8120, Oct. 2020.

[9] N. Pastick, B. Wylie, and Z. Wu, "Spatiotemporal analysis of Landsat-8 and Sentinel-2 data to support monitoring of dryland ecosystems," *Remote Sens.*, vol. 10, no. 5, p. 791, May 2018.

[10] X. Ling and R. Cao, "A new spatiotemporal data fusion method to reconstruct high-quality landsat ndvi time-series data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 2564–2567.

[11] X. Li, L. Wang, Q. Cheng, P. Wu, W. Gan, and L. Fang, "Cloud removal in remote sensing images using nonnegative matrix factorization and error correction," *ISPRS J. Photogramm. Remote Sens.*, vol. 148, pp. 103–113, Feb. 019.

[12] X. Zhu, F. Cai, J. Tian, and T. K.-A. Williams, "Spatiotemporal fusion of multisource remote sensing data: Literature survey, taxonomy, principles, applications, and future directions," *Remote Sens.*, vol. 10, no. 4, p. 527, 2018.

[13] F. Gao, J. Masek, M. Schwaller, and F. Hall, "On the blending of the Landsat and MODIS surface reflectance: Predicting daily Landsat surface reflectance," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 8, pp. 2207–2218, Aug. 2006.

[14] X. Zhu, J. Chen, F. Gao, X. Chen, and J. G. Masek, "An enhanced spatial and temporal adaptive reflectance fusion model for complex heterogeneous regions," *Remote Sens. Environ.*, vol. 114, no. 11, pp. 2610–2623, 2010.

[15] B. Zhukov, D. Oertel, F. Lanzl, and G. Reinhackel, "Unmixing-based multisensor multiresolution image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 3, pp. 1212–1226, May 1999.

[16] M. Lu, J. Chen, H. Tang, Y. Rao, P. Yang, and W. Wu, "Land cover change detection by integrating object-based data blending model of Landsat and MODIS," *Remote Sens. Environ.*, vol. 184, pp. 374–386, Oct. 2016.

[17] A. Li, Y. Bo, Y. Zhu, P. Guo, J. Bi, and Y. He, "Blending multi-resolution satellite sea surface temperature (SST) products using Bayesian maximum entropy method," *Remote Sens. Environ.*, vol. 135, no. 4, pp. 52–63, 2013.

[18] H. Shen, X. Meng, and L. Zhang, "An integrated framework for the spatio–temporal–spectral fusion of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7135–7148, Sep. 2016.

[19] X. Zhu, E. H. Helmer, F. Gao, D. Liu, J. Chen, and M. A. Lefsky, "A flexible spatiotemporal method for fusing satellite images with different resolutions," *Remote Sens. Environ.*, vol. 172, pp. 165–177, Jan. 2016.

[20] Q. Wang and P. M. Atkinson, "Spatio-temporal fusion for daily Sentinel-2 images," *Remote Sens. Environ.*, vol. 204, pp. 31–42, Jan. 2018.

[21] Z. Yin, P. Wu, G. M. Foody, Y. Wu, and F. Ling, "Spatiotemporal fusion of land surface temperature based on a convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1808–1822, Feb. 2020.

[22] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[23] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[24] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[25] C. Ledig *et al.*, "Photo-realistic single image super-resolution using a generative adversarial network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4681–4690.

[26] X. Wang *et al.*, "ESRGAN: Enhanced super-resolution generative adversarial networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV) Workshops*, 2018, pp. 63–79.

[27] J. Cai, H. Zeng, H. Yong, Z. Cao, and L. Zhang, "Toward real-world single image super-resolution: A new benchmark and a new model," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 3086–3095.

[28] H. Song, Q. Liu, G. Wang, L. Hang, and B. Huang, "Spatiotemporal satellite image fusion using deep convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 821–829, Mar. 2018.

[29] D. Jia, C. Cheng, C. Song, S. Shen, L. Ning, and T. Zhang, "A hybrid deep learning-based spatiotemporal fusion method for combining satellite images with different resolutions," *Remote Sens.*, vol. 13, no. 4, p. 645, Feb. 2021.

[30] H. Zhang, Y. Song, C. Han, and L. Zhang, "Remote sensing image spatiotemporal fusion using a generative adversarial network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4273–4286, May 2021.

[31] Z. Tan, M. Gao, X. Li, and L. Jiang, "A flexible reference-insensitive spatiotemporal fusion model for remote sensing images using conditional generative adversarial network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.

[32] Z. Shao, J. Cai, P. Fu, L. Hu, and T. Liu, "Deep learning-based fusion of Landsat-8 and Sentinel-2 images for a harmonized surface reflectance product," *Remote Sens. Environ.*, vol. 235, Dec. 2019, Art. no. 111425.

[33] X. Liu, C. Deng, J. Chanussot, D. Hong, and B. Zhao, "StfNet: A two-stream convolutional neural network for spatiotemporal image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6552–6564, Sep. 2019.

[34] Y. Li, J. Li, L. He, J. Chen, and A. Plaza, "A new sensor bias-driven spatio-temporal fusion model based on convolutional neural networks," *Sci. China Inf. Sci.*, vol. 63, no. 4, Apr. 2020, Art. no. 140302.

[35] D. Jia, C. Song, C. Cheng, S. Shen, L. Ning, and C. Hui, "A novel deep learning-based spatiotemporal fusion method for combining satellite images with different resolutions using a two-stream convolutional neural network," *Remote Sens.*, vol. 12, no. 4, p. 698, Feb. 2020.

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[36] C. Shang *et al.*, "Spatiotemporal reflectance fusion using a generative adversarial network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.
[37] J. Chen, L. Wang, R. Feng, P. Liu, W. Han, and X. Chen, "CycleGAN-STF: Spatiotemporal fusion via CycleGAN-based image generation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5851–5865, Jul. 2021.
[38] A. Vaswani *et al.*, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.
[39] A. Dosovitskiy *et al.*, "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[40] Z. Liu *et al.*, "Swin transformer: Hierarchical vision transformer using shifted Windows," 2021, *arXiv:2103.14030*.
[41] W. Li, D. Cao, Y. Peng, and C. Yang, "MSNet: A multi-stream fusion network for remote sensing spatiotemporal fusion based on transformer and convolution," *Remote Sens.*, vol. 13, no. 18, p. 3724, Sep. 2021.
[42] W.-S. Lai, J.-B. Huang, N. Ahuja, and M.-H. Yang, "Deep Laplacian pyramid networks for fast and accurate super-resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 624–632.
[43] Z. Wang, E. P. Simoncelli, and A. C. Bovik, "Multiscale structural similarity for image quality assessment," in *Proc. 37th Asilomar Conf. Signals, Syst. Comput.*, vol. 2, Nov. 2003, pp. 1398–1402.
[44] I. V. Emelyanova, T. R. McVicar, T. G. Van Niel, L. T. Li, and A. I. J. M. van Dijk, "Assessing the accuracy of blending Landsat–MODIS surface reflectances in two landscapes with contrasting spatial and temporal dynamics: A framework for algorithm selection," *Remote Sens. Environ.*, vol. 133, pp. 193–209, Jun. 2013.
[45] J. Li, Y. Li, L. He, J. Chen, and A. Plaza, "Spatio-temporal fusion for remote sensing data: An overview and new benchmark," *Sci. China Inf. Sci.*, vol. 63, no. 4, pp. 1–17, Apr. 2020.
[46] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.
[47] Z. Wang and A. C. Bovik, "A universal image quality index," *IEEE Signal Process. Lett.*, vol. 9, no. 3, pp. 81–84, Mar. 2002.
[48] R. H. Yuhas, A. F. Goetz, and J. W. Boardman, "Discrimination among semi-arid landscape endmembers using the spectral angle mapper (SAM) algorithm," in *Proc. Summaries 3rd Annu. JPL Airborne Geosci. Workshop*, vol. 1, 1992, pp. 147–149.
[49] M. M. Khan, L. Alparone, and J. Chanussot, "Pansharpening quality assessment using the modulation transfer functions of instruments," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 11, pp. 3880–3891, Nov. 2009.
[50] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1251–1258.



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